

Executive Summary — Surrogate Model Validation Report

Use case: UCFATIGUE_1 | 28 July 2026 | 2 model(s) evaluated

REQUIREMENTS PARTIALLY MET

Recommended model: GradientBoosting

Avg R²: 0.9643 — Excellent

Q90 Accuracy — Partially met

Pass: 1g, Vertical maneuver, Vertical gust, Turn, n0

Fail: Frontal gust, Giro

Overfitting check (KS) — Overfitting present

Pass: Vertical maneuver, Frontal gust

Fail: 1g, Vertical gust, Turn, n0, Giro

Project Overview

Use case	UCFATIGUE_1
Inputs	FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)
Outputs	7: 1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro
Train / Val / Test	80% / 10% / 10%
Accuracy target	Q90 < 10% relative error per output

Model Performance — GradientBoosting (Test Set)

Output	R ²	Q90	Target	Gap	Status
1g	0.9797	0.0257	10%	-0.0743	PASS
Vertical maneuver	0.9723	0.0318	10%	-0.0682	PASS
Vertical gust	0.9914	0.0262	10%	-0.0738	PASS
Turn	0.9286	0.0351	10%	-0.0649	PASS
Frontal gust	0.9964	0.2795	10%	+0.1795	FAIL
n0	0.9317	0.0149	10%	-0.0851	PASS
Giro	0.9499	0.2215	10%	+0.1215	FAIL

Q90 < 10%

Q90 ≥ 10%

R² ≥ 0.95

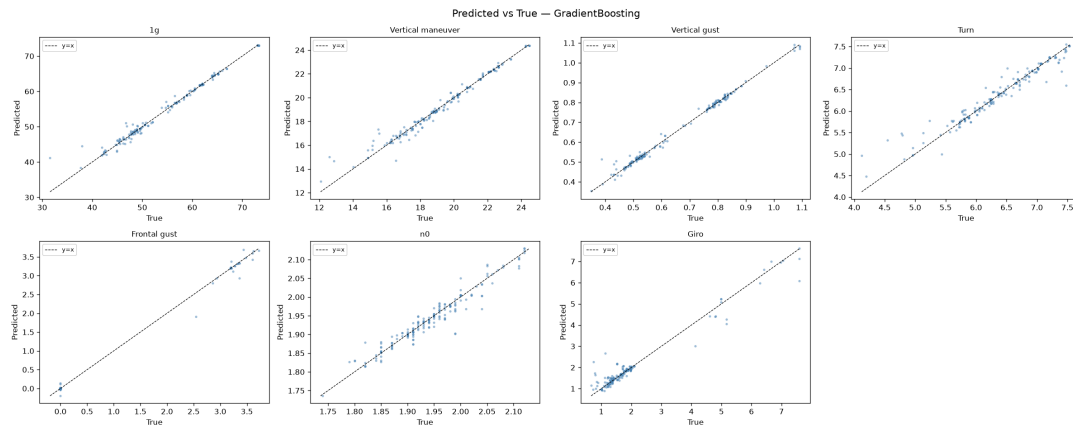
R²

∈ [0.80, 0.95) Gap = Q90−target (< 0 ⇒ margin)

Analysis & Recommendations

- Significant failures (Frontal gust, Giro): Q90 exceeds target by ≥ 5%. Investigate input features and model capacity.
- Overfitting detected (1g, Vertical gust, Turn, n0, Giro): KS test rejects equal residual distributions on train and test.

Predicted vs True — GradientBoosting (Test Set, up to 5 000 points)



Overfitting Check — KS Test p-values (GradientBoosting)

H_0 : residuals on train and test follow the same distribution. $p \geq 0.05$ no overfitting. $p < 0.05$ overfitting.

Output	KS p-value
1g	0.004
Vertical maneuver	0.161
Vertical gust	0.042
Turn	0.044
Frontal gust	0.974
n0	0.032
Giro	0.017

Full technical details in the [Technical Review](#) (page 5).

Model Comparison — UCFATIGUE_1

28 July 2026

Overall Summary

Model	Avg R ²	Q90 pass (7 outputs)	KS pass (7 outputs)	★ Recommended model. Q90 target: < 10%. KS: $p \geq 0.05$ = no overfitting.
MLP	0.5237	5/7	7/7	
GradientBoosting ★	0.9643	5/7	2/7	

Per-Output Metrics (all models)

Output	MLP			GradientBoosting ★			
	R ²	Q90	Gap	R ²	Q90	Gap	
lg	0.9160	0.0473	-0.0527	0.9797	0.0257	-0.0743	R² ≥ 0.95 R² ∈ [0.80, 0.95)
Vertical maneuver	0.9116	0.0554	-0.0446	0.9723	0.0318	-0.0682	
Vertical gust	0.9377	0.0918	-0.0082	0.9914	0.0262	-0.0738	
Turn	0.7544	0.0783	-0.0217	0.9286	0.0351	-0.0649	
Frontal gust	0.9794	11.3300	+11.2300	0.9964	0.2795	+0.1795	
n0	-0.7292	0.0723	-0.0277	0.9317	0.0149	-0.0851	
Giro	-0.1042	0.9005	+0.8005	0.9499	0.2215	+0.1215	

Fail Gap = Q90−target

Technical Review

UCFATIGUE_1 — Full Validation Report

Technical Review — UCFATIGUE_1

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Introduction

This report documents the full validation of surrogate models for use case UCFATIGUE_1. 2 model architecture(s) were trained and evaluated: MLP, GradientBoosting. Each model predicts 7 structural output quantities from 8 input features (FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)). The accuracy criterion requires the 90th-percentile relative error (Q90) to remain below 10% for every output on unseen test data.

Model Selection

GradientBoosting achieved the highest average R^2 on the test set (0.9643), outperforming MLP ($R^2=0.5237$). It is the recommended model for deployment.

Accuracy Assessment (Q90 & R^2)

GradientBoosting satisfies Q90 on 5/7 outputs. Passing: 1g, Vertical maneuver, Vertical gust, Turn, n0. Failing: **Frontal gust, Giro**. Refer to the Improvement Roadmap for targeted corrective actions.

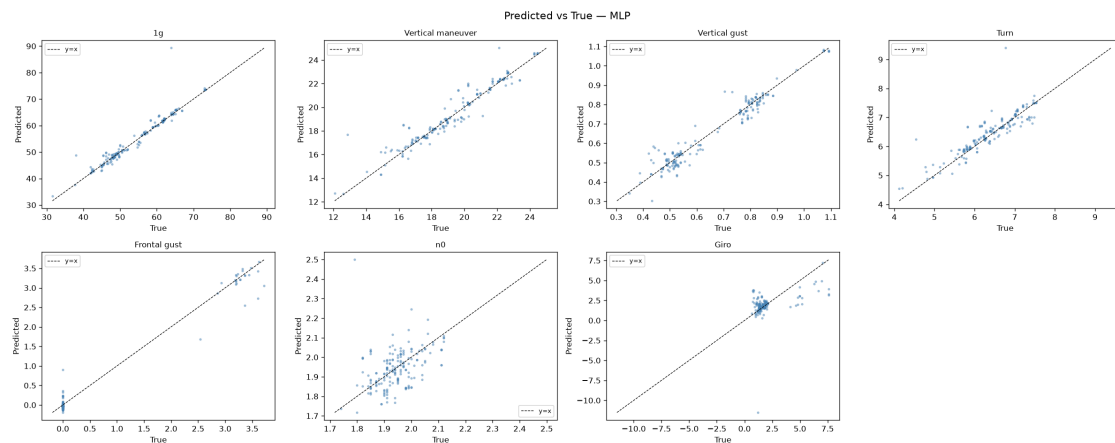
Output	MLP			GradientBoosting ★		
	R^2	Q90	Gap	R^2	Q90	Gap
1g	0.9160	0.0473	-0.0527	0.9797	0.0257	-0.0743
Vertical maneuver	0.9116	0.0554	-0.0446	0.9723	0.0318	-0.0682
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Giro	-0.1042	0.9005	+0.8005	0.9499	0.2215	+0.1215

Fail Gap = Q90–target (< 0 $\Rightarrow$ margin)

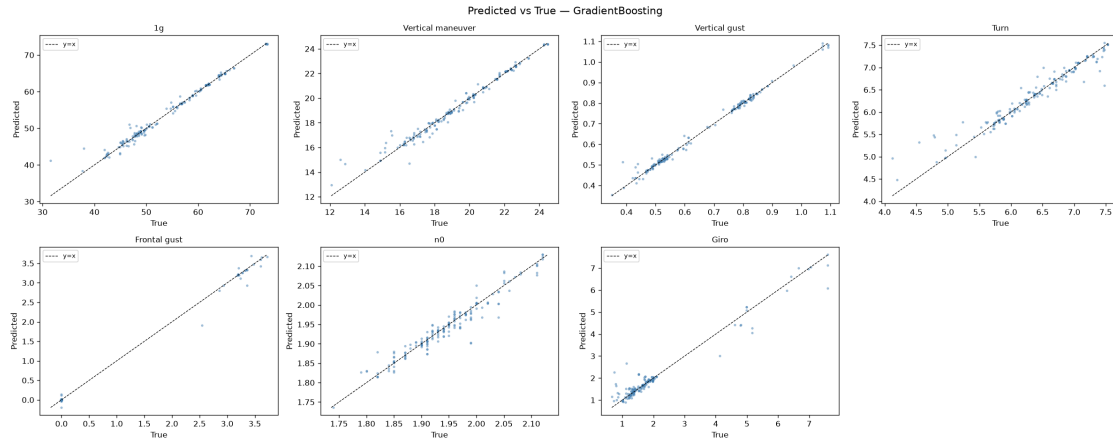
Predicted vs True (Scatter Plots, Test Set)

Each plot shows up to 5 000 test-set points. The dashed diagonal represents perfect prediction ($\hat{y} = y$). Systematic bias or fan-shaped spread (heteroscedasticity) indicate model limitations.

MLP



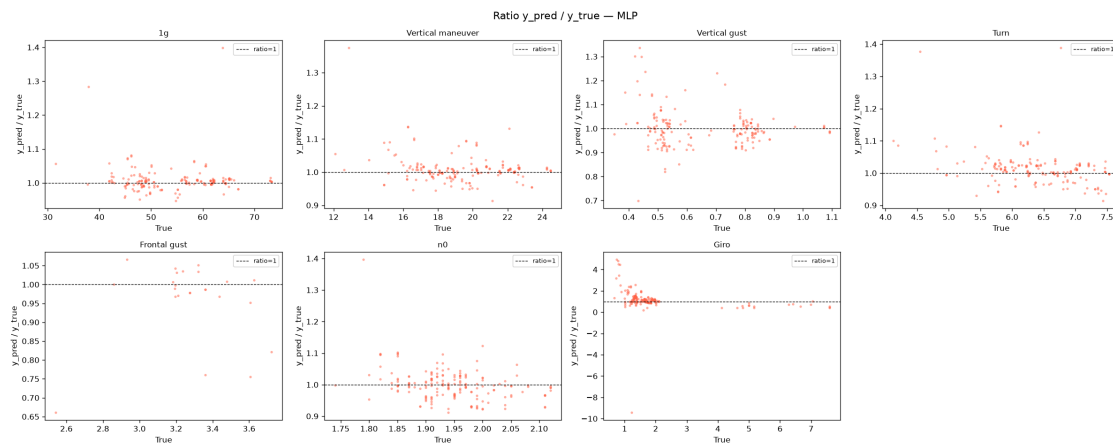
GradientBoosting ★



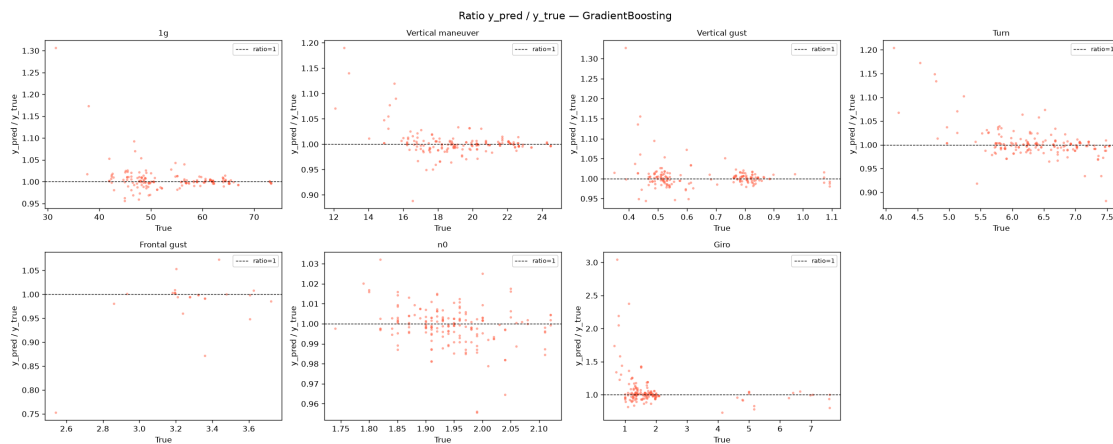
Relative Error Analysis (Ratio Plots, Test Set)

Each panel shows the ratio $\hat{y}/y$ as a function of the true value. A ratio of 1.0 (dashed line) corresponds to perfect prediction. Systematic deviation from 1.0 reveals bias; increasing spread with magnitude reveals heteroscedasticity.

MLP



GradientBoosting ★



Overfitting Assessment (KS Test)

The Kolmogorov–Smirnov test compares residual distributions on train vs test sets. H_0 : both distributions are identical (no overfitting). $p < 0.05$ is statistical evidence of overfitting.

The KS test detects distributional differences for GradientBoosting on: **1g, Vertical gust, Turn, n0, Giro**. This suggests localised overfitting on these outputs. Increasing regularisation or reducing model capacity is recommended.

Output	MLP	GradientBoosting
1g	0.473	0.004
Vertical maneuver	0.180	0.161
Vertical gust	0.635	0.042
Turn	0.078	0.044
Frontal gust	0.736	0.974
n0	0.378	0.032
Giro	0.781	0.017

$p \geq 0.05$ no overfitting $p < 0.05$ overfitting detected

Data Split Quality (VTP Analysis)

The Voronoi Tesselation Proximity method verifies whether test points lie within the training distribution. A residual voxel proportion ≤ 0.05 and chi-squared $p \geq 0.05$ confirm a high-quality, unbiased split. The VTP analysis confirms the test set is well-distributed within the training space (residual voxel proportion ≤ 0.05 , no isolated test points). The chi-squared test validates the split is statistically unbiased. Train/test partition is of high quality.

Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.931	
Phacking test proportion	0.006	
Isolated test proportion	0.063	
Chi ² p-value (≥ 0.05)	0.7969	✓

Improvement Roadmap

Prioritised actions based on the failure analysis above.

- Significant failures (Frontal gust, Giro): Q90 exceeds target by $\geq 5\%$. Investigate input features and model capacity. Review whether additional physical drivers are missing from the input feature set, or consider a more expressive architecture (deeper network, ensemble).
- Overfitting detected (1g, Vertical gust, Turn, n0, Giro): KS test rejects equal residual distributions on train and test. Increase regularisation (dropout, weight decay, early-stopping patience) or reduce model capacity.